U4B

Upcycle 4 Better — Fabric Donation & Recycling Platform

Status: Live in Production • Platform: Web Application • Market: Malaysia

U4B (Upcycle 4 Better) is a full-stack web application designed to digitize and streamline the fabric donation and recycling rewards process across Malaysia. The platform connects donors with 50+ recycling bin locations, enabling verified donations and rewarding users with vouchers from partner stores like Zalora and Best Bundle.

The Problem

Before U4B, the fabric donation and verification process was entirely manual, creating friction for both donors and administrators:

For Donors

- Had to physically visit bins with no way to confirm bin locations or availability
- Required recording a video of their donation on their personal phone
- Had to manually email the video to U4B employees for verification
- Waited days for manual review before receiving voucher codes
- No way to track donation history or accumulated rewards

For U4B Administrators

- Received donation videos via email — no centralized system
- Had to manually watch each video to verify legitimate donations
- No way to verify if the donor was actually at the bin location
- Manually generated and sent voucher codes via email reply
- No analytics or tracking of donation volumes across bins
- Prone to fraudulent claims with no location verification

The Core Issues

- Time-consuming: Average 3-5 days from donation to voucher delivery
- No accountability: Impossible to verify donor was at actual bin
- Scattered data: Donation records spread across email threads
- Poor user experience: Multi-step process discouraged repeat donations

The Solution

I built a comprehensive web application that digitizes the entire donation workflow — from bin discovery to voucher redemption — with built-in verification and fraud prevention.

For Donors (Mobile-Optimized Web App)

- Account Creation: Simple phone/email registration with secure authentication
- Bin Discovery: Interactive map showing all 50+ bin locations across Malaysia
- QR Scanning: Scan QR code on any bin to automatically register the location
- In-App Recording: Record donation video directly within the application
- Geolocation Verification: GPS coordinates captured automatically during recording
- Instant Submission: One-tap submission with automatic location tagging

- Reward Tracking: View donation history, pending verifications, and earned vouchers
- Voucher Redemption: Access and redeem vouchers directly from partner stores

For Administrators (Web Dashboard)

- Centralized Queue: All pending donations in one organized dashboard
- Location Verification: See donor's GPS distance from the scanned bin
- Video Review: Watch donation videos directly in the browser
- One-Click Approval: Approve or reject with automated voucher generation
- Fraud Detection: Flag donations where GPS doesn't match bin location
- Analytics Dashboard: Track donations by bin, region, and time period
- User Management: View donor profiles, history, and account status

How It Works

Donor Flow

- 1. Open U4B app and navigate to 'Donate' section
- 2. Find nearest bin on map or scan QR code at physical bin
- 3. App captures GPS location and identifies the specific bin
- 4. Record video of donation (clothes going into bin)
- 5. Submit — video + location sent for verification
- 6. Receive push notification when donation is verified
- 7. Voucher automatically added to account for redemption

Admin Verification Flow

- 1. New donation appears in pending queue with location data
- 2. System shows distance between donor GPS and bin coordinates
- 3. Admin reviews video to confirm legitimate donation
- 4. If location matches and video is valid → Approve
- 5. System auto-generates voucher and notifies donor
- 6. If suspicious → Reject with reason or flag for review

Technical Architecture

Frontend

- Framework: Next.js 16 with React 19
- Language: TypeScript (98% type coverage)
- Styling: Tailwind CSS for responsive, mobile-first design
- Icons: Lucide React for consistent iconography

Backend

- Runtime: Node.js with Express 5
- Database: PostgreSQL for relational data
- Authentication: JWT with bcrypt password hashing
- File Uploads: Multer for video handling
- API Design: RESTful architecture with proper error handling

Security Features

- JWT-based authentication with secure token refresh

- Role-based access control (RBAC) — User vs Admin
- Bcrypt password hashing with salt rounds
- SQL injection prevention via parameterized queries
- HTTPS encryption for all data transmission

Deployment

- Frontend & Backend: Firebase Hosting
- Database: Cloud PostgreSQL instance
- File Storage: Cloud storage for video files

Results & Impact

- Deployment: Live across 50+ bin locations in Malaysia
- Verification Time: Reduced from 3-5 days to under 24 hours
- Fraud Prevention: GPS verification eliminated false claims
- User Experience: Single app replaces email + phone + waiting
- Admin Efficiency: Centralized dashboard replaces scattered emails
- Data Insights: First-ever analytics on donation patterns by location

My Role

I was the sole developer on this project during my internship at Upcycle 4 Better (Sep 2025 — Jan 2026). I independently handled:

- Full system architecture and database design
- Complete frontend development (Next.js, TypeScript, Tailwind)
- Complete backend development (Node.js, Express, PostgreSQL)
- Authentication system with JWT and role-based access
- QR code scanning and geolocation integration
- Video upload and verification workflow
- Admin dashboard with analytics
- UI/UX design based on brand guidelines
- Deployment and production configuration

This case study represents a live production system.